Task 2: Basic Investigation Process

1. Introduction

Digital forensic investigation is a systematic process used to identify, preserve, analyze, and present digital evidence in a legally admissible manner. With the increasing dependency on digital systems, cyber incidents such as data theft, unauthorized access, and malware attacks have become common. Therefore, a structured investigation process is essential to ensure that digital evidence is handled in a way that maintains its integrity and credibility.

This report provides a detailed explanation of the digital forensic investigation process along with a practical simulation of recovering deleted files from a storage device.

2. Objectives of the Study

The key objectives of this task are:

- To understand the **complete lifecycle of a digital forensic investigation**
- To apply a **step-by-step methodology**
- To ensure **data integrity using forensic techniques**
- To gain hands-on understanding of forensic tools such as Autopsy and FTK Imager
- To document findings in a **professional forensic report format**

3. Overview of Investigation Methodology

Digital forensic investigations follow a standardized methodology to ensure consistency and reliability. The process can be summarized into five major phases:

1. Identification
2. Preservation
3. Analysis
4. Documentation
5. Presentation

Each phase is interdependent and must be executed carefully.

4. Detailed Phases of Investigation

4.1 Identification Phase

This is the initial phase where potential digital evidence is recognized and collected.

Key Activities:

- Identifying digital devices (USB drives, laptops, servers)
- Recognizing relevant data sources (files, logs, applications)
- Defining investigation scope and objectives

Technical Considerations:

- Volatile vs non-volatile data
- Live systems vs powered-off devices

Example in This Case:
A USB drive suspected of containing deleted confidential files is identified as the primary evidence.

4.2 Preservation Phase

The preservation phase ensures that the original evidence remains unchanged throughout the investigation.

Key Activities:

- Creating a forensic image (bit-by-bit copy)
- Using write blockers to prevent modification
- Generating cryptographic hash values

Hashing Concept:
Hashing ensures integrity using algorithms like SHA-256.

- Before analysis → hash generated
- After analysis → hash verified

If both hashes match → evidence is unchanged

Tools Used:

- FTK Imager
- HashMyFiles

4.3 Analysis Phase

This is the most critical phase where evidence is examined.

Key Activities:

- File recovery (deleted files)
- Keyword searching
- Metadata analysis
- Timeline reconstruction
- File system examination (NTFS/FAT)

Tools Used:

- Autopsy

Technical Insights:

- Deleted files are not immediately erased; only references are removed
- Forensic tools scan disk sectors to recover such files

4.4 Documentation Phase

Documentation is crucial for maintaining transparency and reproducibility.

Key Activities:

- Recording every action performed
- Logging timestamps
- Listing tools and versions used
- Capturing screenshots of findings

Importance:

- Helps in legal proceedings
- Ensures investigation can be repeated
- Provides audit trail

4.5 Presentation Phase

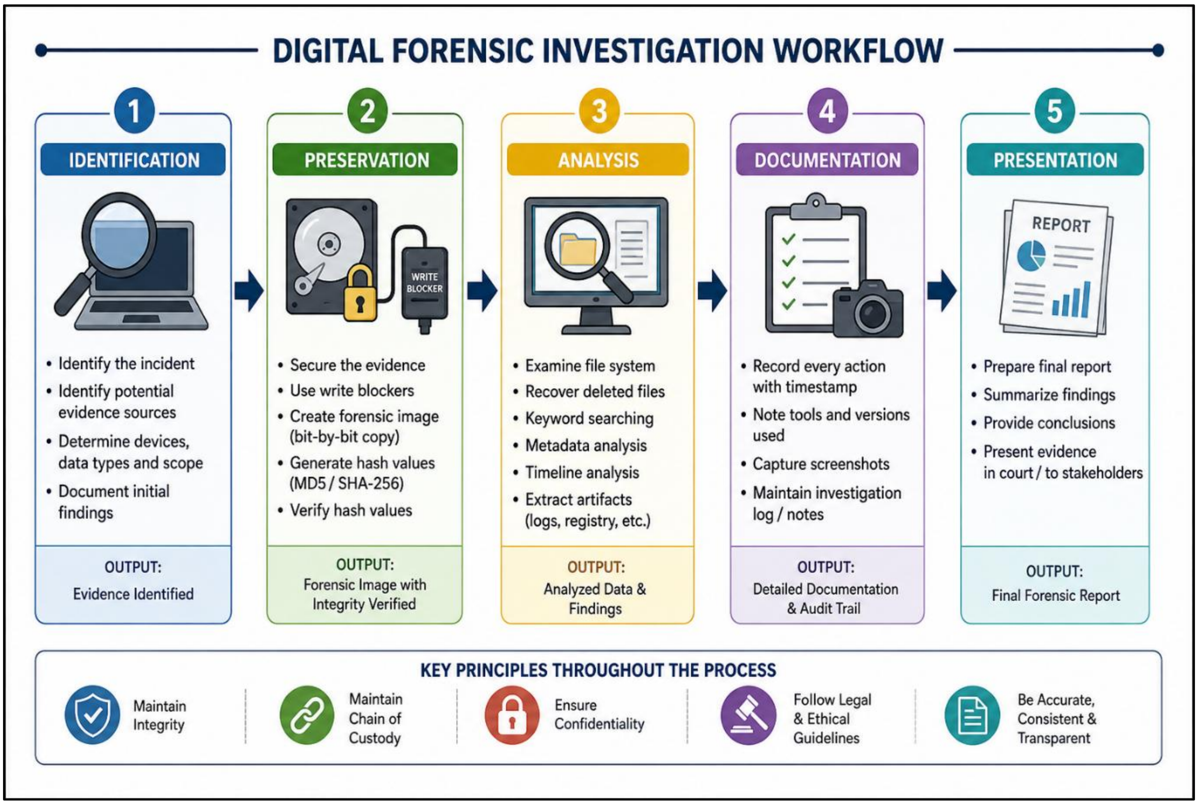
The final phase involves communicating findings in a structured format.

Key Activities:

- Writing final forensic report
- Summarizing findings

- Providing conclusions
- Supporting with evidence (screenshots, logs)

5. Investigation Workflow



6. Practical Implementation

Case Title:

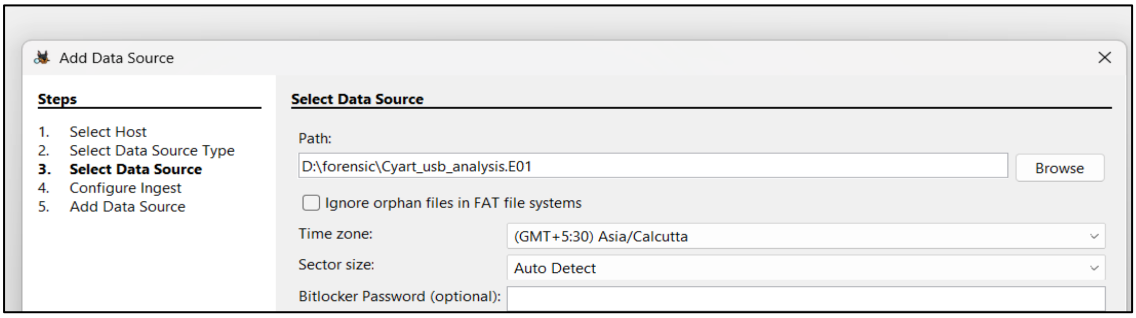
Recovery of Deleted Files from USB Drive

Case Description:

A USB drive was suspected of containing deleted confidential documents. The objective was to recover and analyze the deleted data without altering the original evidence.

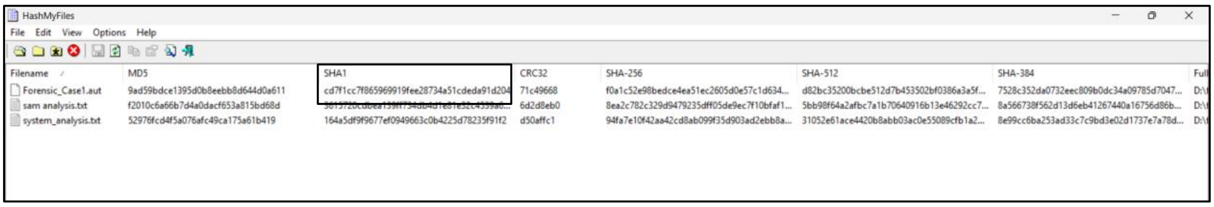
Step 1: Identification

- Device identified: USB Drive (16GB)
- Type: Removable storage
- Objective: Recover deleted files



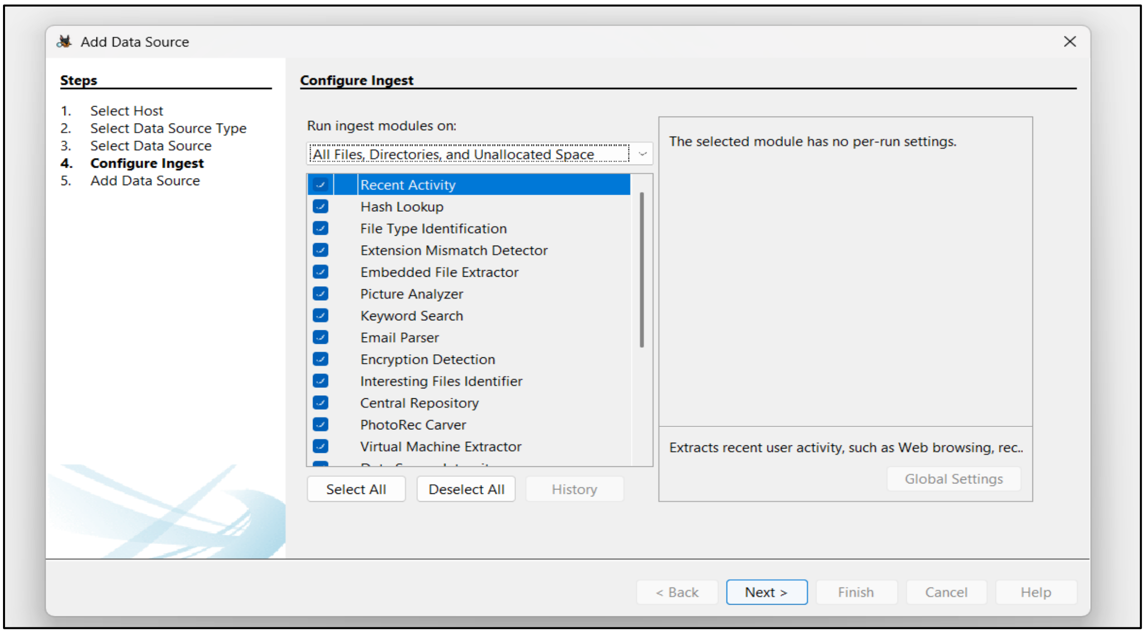
Step 2: Preservation

- USB connected safely
- Hash value generated.
- Verified that hash remains same after analysis



Step 3: Analysis

- Evidence analyzed using **Autopsy**



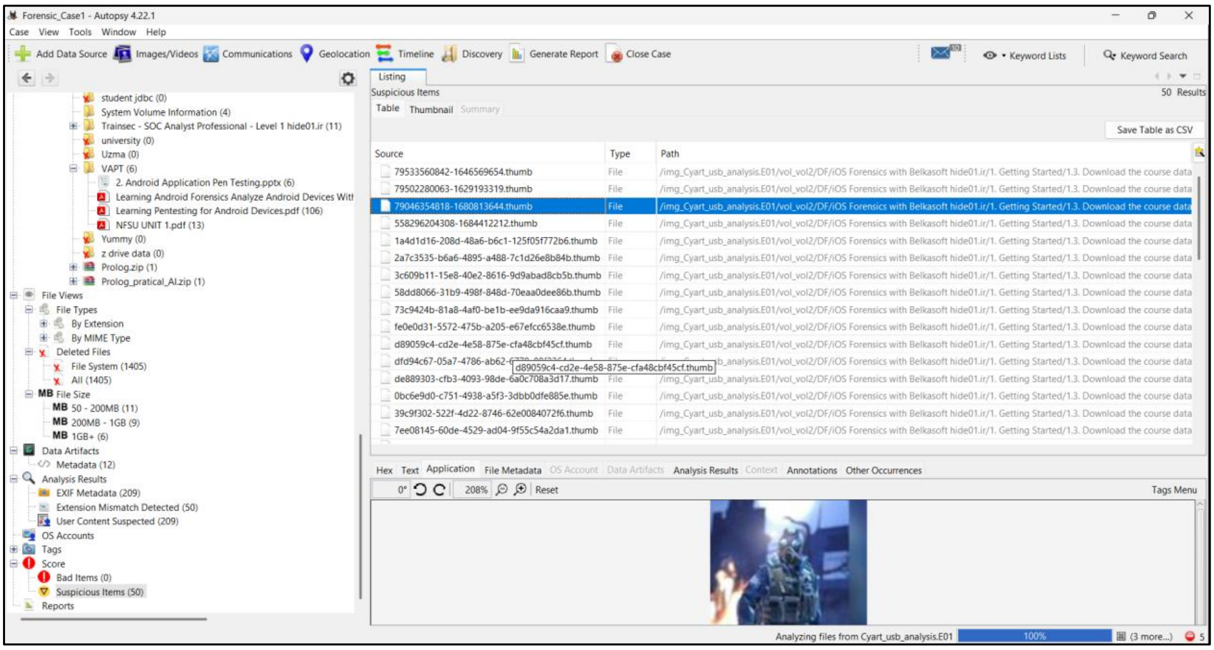
Steps Performed:

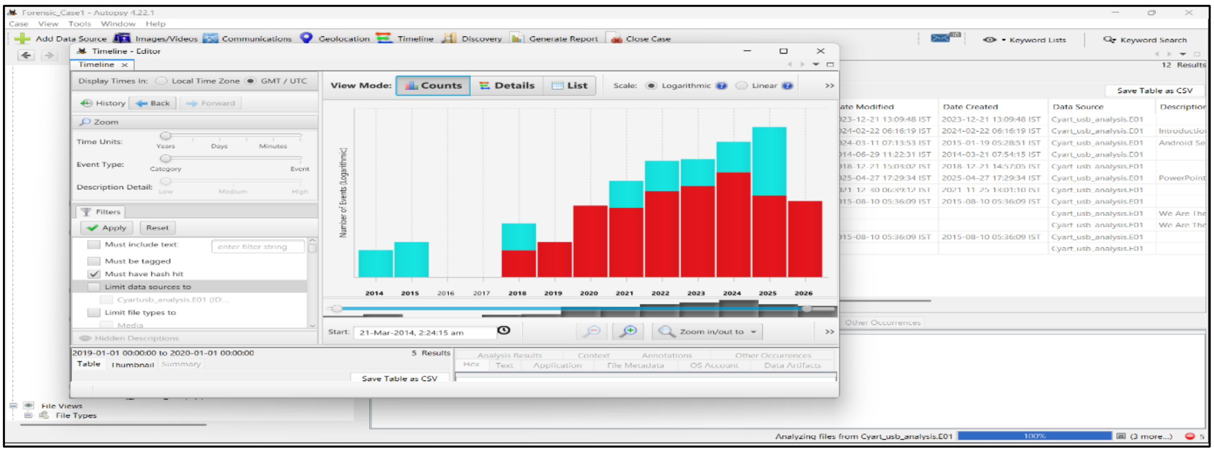
- Loaded disk image

- Performed file system scan
- Enabled file recovery module
- Searched for deleted files

Findings:

- 3 deleted files recovered
- File types: PDF, DOCX
- One file contained confidential business data

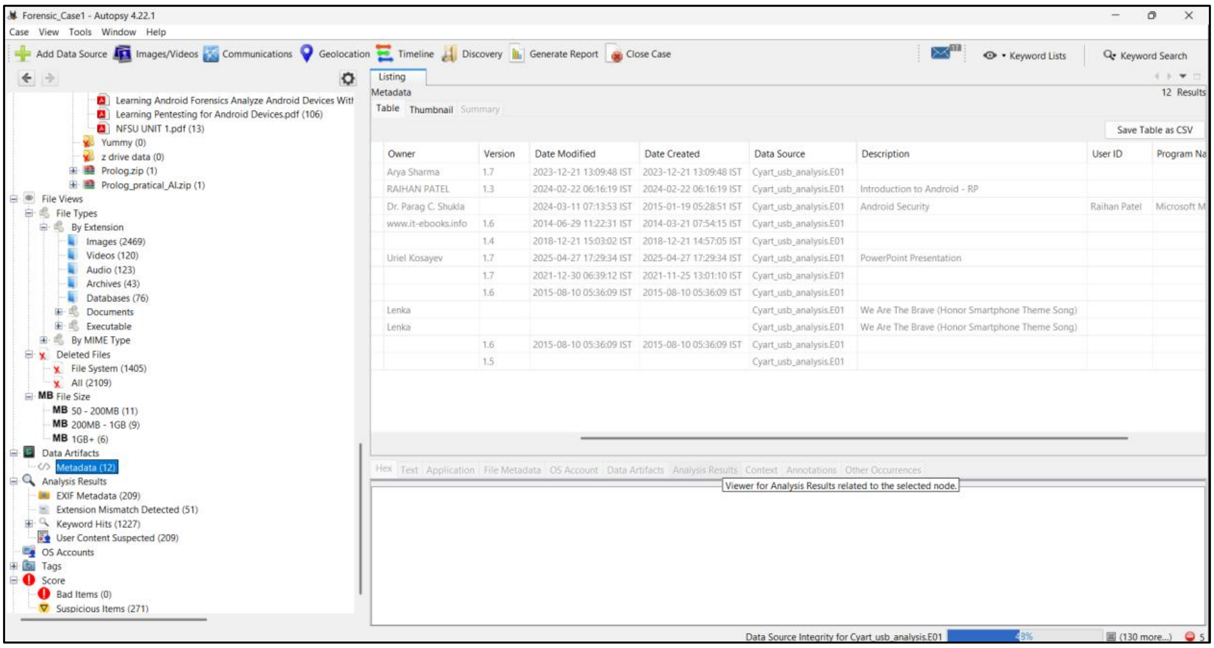




Step 4: Documentation

- All actions recorded step-by-step

- Tools used documented
- Screenshots captured
- Timeline created



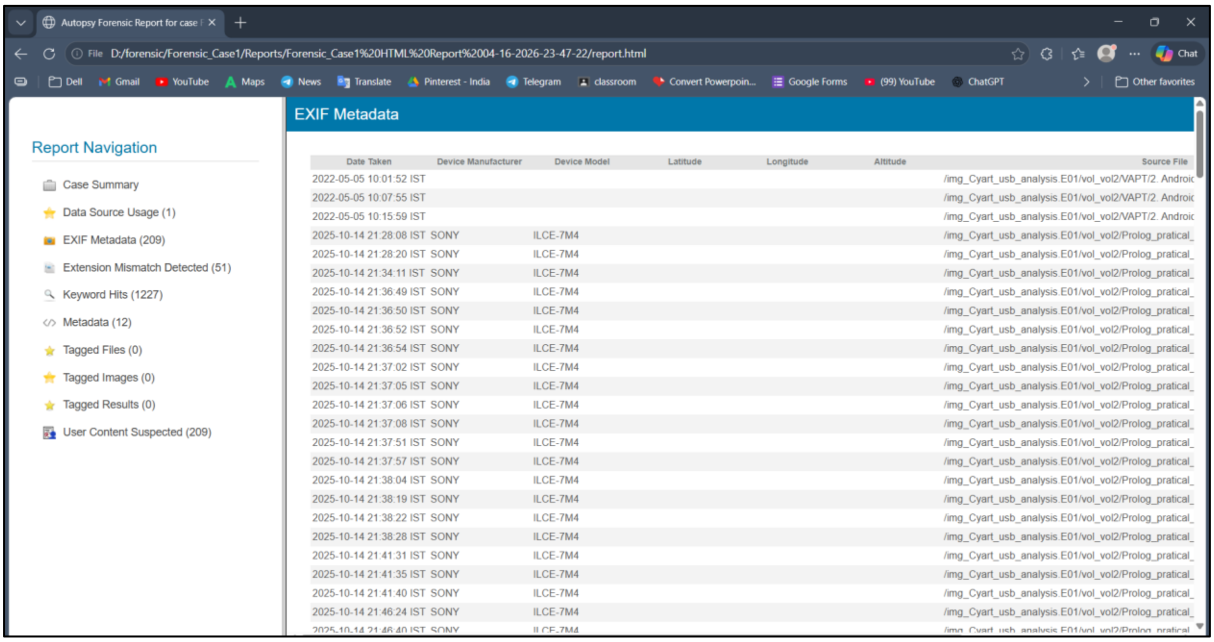
Step 5: Presentation

Final Findings:

- Deleted files successfully recovered
- Evidence confirms presence of sensitive information

Conclusion:

- Data deletion was attempted but not fully removed
- Evidence remains recoverable and valid



7. Tools Used in Investigation

- Autopsy – File recovery and analysis
- FTK Imager – Disk imaging
- HashMyFiles – Integrity verification

8. Challenges Faced

- Handling large data volumes
- Identifying relevant files
- Understanding file system structure
- Ensuring no accidental modification

9. Best Practices

- Always create forensic image before analysis
- Never work on original evidence
- Maintain proper documentation
- Verify integrity using hash values
- Follow chain of custody

10. Importance of Investigation Process

- Ensures **legal admissibility**
- Maintains **evidence integrity**
- Provides **structured approach**
- Enhances **credibility of investigation**

11. Conclusion

The digital forensic investigation process provides a well-defined framework for handling digital evidence. By strictly following all phases—Identification, Preservation, Analysis, Documentation, and Presentation—investigators can ensure accurate and reliable results. This task demonstrated both theoretical knowledge and practical implementation, highlighting the importance of a structured forensic approach.